

Unit 1, Lesson 5: Solving One-Step Equations

Benchmarks

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.

MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.

Learning Target

- ☐ I can solve one-step equations involving addition, subtraction, multiplication, and division using models and inverse operations.

1.5.1 Warm-Up: Brain Capacity

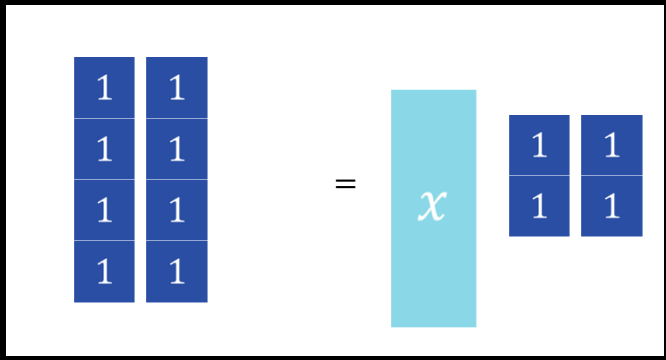
A human brain can store about 3,500 trillion bytes of data, while a cat's brain can store about 98 trillion bytes of data.

1. What is the brain storage capacity of human to cat?
2. About how many brains of cats would it take to store the same amount of data as the human brain?

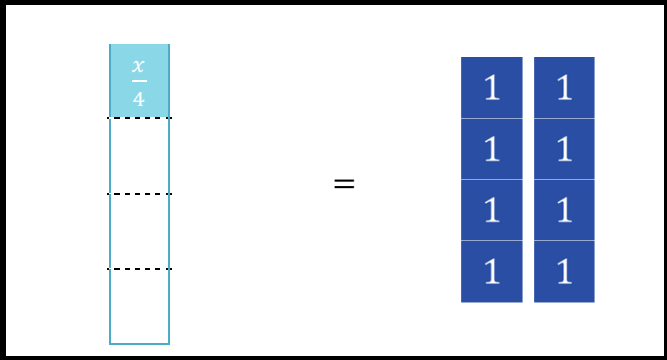
1.5.2 Guided Practice: Solving Equations

1. Four algebraic equations are represented using algebra tiles.

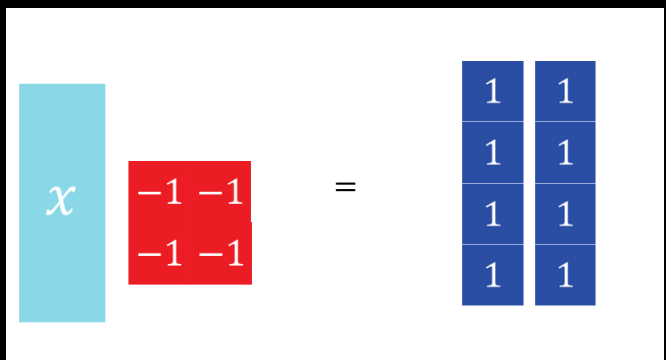
Model A



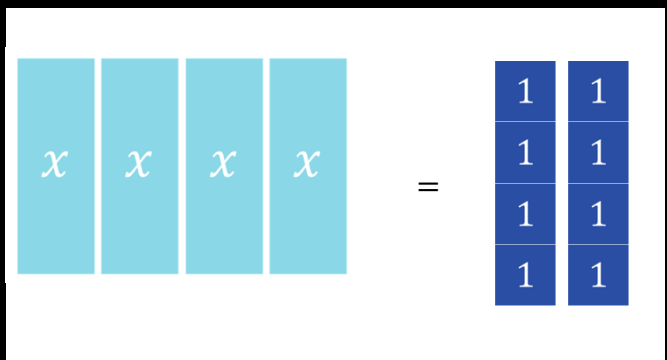
Model B



Model C



Model D



a. Match the algebra tile model to the equations.

Equation	Model
$4x = 8$	
$\frac{x}{4} = 8$	
$8 = x + 4$	
$x - 4 = 8$	

b. Use the algebra tile models to determine the value of x in each equation.

Equation	Value of x
$4x = 8$	
$\frac{x}{4} = 8$	
$8 = x + 4$	
$x - 4 = 8$	

- c. Choose one of the equations and explain how you determined the value of x from the model.

When finding the solution to an equation, **inverse operations** are used to balance an equation.

An **inverse operation** is an operation that reverses the effect of another operation.

2. Using the definition of inverse operations, complete the table to indicate which operation would reverse the given operation.

Operation	Inverse Operation
Subtraction (−3)	
Addition (+6)	
Multiplication (× 1.5)	
Division (÷ 10)	

3. Use the equation $21 = 7h$ to complete the following.
- a. What mathematical operation is represented by $7h$?

-
- b. What is the inverse operation?
- c. Perform the inverse operation on both sides of the equation. Show your work.
 $21 = 7h$
- d. What is the value of h ?
- e. Discuss with your partner how to check whether or not your solution is correct. Then, check your solution.
4. Use the equation $\frac{b}{-4} = 13$ to complete the following:
- a. What mathematical operation is represented by $\frac{b}{-4}$?
- b. What is the inverse operation?
- c. Perform the inverse operation on both sides of the equation. Show your work.
 $\frac{b}{-4} = 13$

- d. What is the value of b ?
- e. Verify your solution is correct.

1.5.3 Collaboration: Choose a Strategy

1. An elevator starts in the parking garage under a building at level -3 , which is 3 levels underground. The elevator travels to the sixth floor of the building. This situation can be modeled by the equation $-3 + x = 6$, where x represents the number of floors the elevator travels.
- Tyler says that since subtraction is the inverse operation of addition, you should use subtraction to solve. Eva says that you should add 3 to both sides of the equation to solve it.

a. Complete the table by showing the result of each student’s strategy.

Tyler’s Strategy	Eva’s Strategy
$(-3) + x = 6$	$(-3) + x = 6$
$x = \underline{\hspace{2cm}}$	$x = \underline{\hspace{2cm}}$

- b. With your partner, determine whose strategy is correct. Explain how you know.

1.5.4 Try It: Solving Equations

1. Each equation shown in the table can be solved in one step. Complete the table by describing how to solve each equation, showing your work, and providing the solution. The first row is started for you.

Equation	Description	Work	Solution	Verify
$6x = 18$	divide both sides of the equation by 6			
$h - 5 = -7$				
$-36 = \frac{m}{-6}$				
$32 = 41 + a$				
$-45 = 9y$				

2. Compare your work with your partner and resolve any differences. Verify your solutions using substitution.

1.5.5 Collaboration: Solving Equations

With your partner, determine who will be Partner A and who will be Partner B.

1. Solve the equations from your designated column.

Partner A	Partner B
$72 = -9x$	$x - 4 = -12$
$469 + z = 459$	$\frac{z}{5} = -2$
$3 = \frac{w}{-6}$	$-2w = 36$
$-6 = x - 20$	$-8 = -22 + x$

2. The equations in each row have the same solutions. Check your work with your partner, making revisions to your work if necessary.

1.5.6 Wrap-Up: Reflection

1. Draw an X on each line to indicate where your current learning is related to each target.

☐ I can solve one-step equations involving addition and subtraction.

I need help.

I'm getting there, but

I understand this

I can't get started.

I need more practice.

and I feel confident.



☐ I can one-step equations involving multiplication and division.

I need help.

I'm getting there, but

I understand this

I can't get started.

I need more practice.

and I feel confident.

